Simultaneous Surface Reduction and Self-Passivation via Ar-CH_4 Plasma for Cu Hybrid Bonding

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***Abstract*— As semiconductor scaling reaches its physical limits, Cu hybrid bonding has emerged as a cornerstone of advanced packaging for high-performance computing and AI applications. To achieve reliable low-temperature bonding while preventing Cu re-oxidation, surface passivation is essential. This study proposes a novel single-step Ar-CH_4 plasma treatment using an inductively coupled plasma (ICP) system to simultaneously reduce Cu oxides and deposit an amorphous hydrogenated carbon (a-C:H) passivation layer. The plasma process was optimized through a Central Composite Design (CCD), and the treatment time was varied from 15 to 60 seconds to precisely control the a-C:H layer thickness. Surface analysis via XPS and TEM confirmed the complete removal of Cu^{2+} species and the linear growth of the a-C:H layer. Cross-sectional TEM observations revealed that the a-C:H thickness plays a dual role: at the optimized P15 condition (~2.5 nm), the layer acts as a diffusion-active species, facilitating a synergistic triple-diffusion mechanism that results in a seamless, monolithic Cu interface. In contrast, thicker layers (P60) act as a diffusion barrier, effectively suppressing metallic intermixing. These findings demonstrate that precise nanometer-scale control of the a-C:H layer is critical for enabling robust, void-free Cu-to-Cu bonding, providing a highly scalable solution for next-generation high-density interconnects.**

Keywords—Cu Hybrid bonding, Reduction, Passivation, Ar-CH_4 plasma treatment

This work was supported by the Technology Innovation Program (Public-Private Joint Investment Semiconductor R&D Program [K-CHIPS] for nurturing high-level human resources; RS-2023-00236091, Development of hybrid bonding technologies for 3D packaging interconnects), funded by the Ministry of Trade, Industry and Energy (MOTIE, Korea; 1415187584). Additional support provided by a grant of the Basic Research Program funded by the Korea Institute of Machinery and Materials (grant number: NK254A, Development of Core Technologies for Advanced Chiplet Packaging Equipment). *(Corresponding author: Sarah Eunkyung Kim).*

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I. Introduction

As transistor scaling approaches its physical limits, Advanced Packaging has emerged as a new paradigm to sustain the evolution of semiconductor performance. The industry's technological roadmap is aggressively shifting toward maximizing I/O density and accelerating data rates to meet the massive data transfer requirements of AI-driven computation. While increasing the data rate per channel often encounters challenges like power dissipation and electromagnetic interference (EMI) issues, achieving ultra-high I/O density via advanced interconnects provides a more efficient path for high-bandwidth communication with reduced energy per bit [1-4]. Among various interconnect technologies, such as EMIB, Si interposers, and TSVs, Cu hybrid bonding has demonstrated the greatest potential for sub-micron pitch scaling. By eliminating conventional solder bumps, it significantly reduces parasitic capacitance and enables unprecedented interconnect density. In the hybrid bonding process, plasma treatment is essential not only for surface cleaning and activation but also for passivating the cleaned surface to ensure reliability, particularly in die-to-wafer (D2W) bonding [5-7]. Conventional Ar/N_2 two-step plasma treatments have been employed to remove Cu oxides through Ar physical bombardment, subsequently passivating the surface with a thin Cu_4N layer to prevent immediate re-oxidation [5]. In this study, we propose a novel Ar-CH_4 plasma treatment for the simultaneous reduction and passivation of Cu surfaces in a single-step process. By utilizing H radicals and ions generated within the Ar-CH_4 plasma, we effectively reduce Cu oxides while concurrently forming an amorphous hydrogenated carbon (a-C:H) passivation layer. The exposure of Cu surfaces to hydrogen-based plasma effectively promotes the chemical reduction of native Cu oxides through the active interaction of hydrogen radicals, which is essential for achieving a clean and oxide-free metallic interface [8]. This approach offers significant advantages by streamlining the process and providing robust protection against re-oxidation. Furthermore, we demonstrate that the a-C:H layer can act as either a diffusion activator or a diffusion barrier depending on its nanometer-scale thickness. Consequently, precise thickness control of the a-C:H layer is presented as a vital factor for achieving reliable Cu diffusion bonding and superior interfacial integrity.

II. Method

A. Plasma Treatment and Surface Characterization

The experimental procedure began with the preparation of 8-inch notch-type silicon substrates pre-coated with a 700 nm SiO_2 insulating layer. To ensure robust adhesion, a 50 nm titanium (Ti) adhesion layer and a 1um copper (Cu) thin film were sequentially deposited using DC magnetron sputtering. The sputtering process was performed under a working pressure of 5 mTorr with a DC power of 2000 W and an Ar flow rate of 100 sccm. Subsequent to the metal deposition, the Cu surface was subjected to Ar-CH_4 plasma treatment using an inductively coupled plasma chemical vapor deposition (ICP-CVD) system (Oxford PP100 Cobra300). The plasma treatment conditions were optimized using a Central Composite Design (CCD) approach within the Design of Experiments (DOE) framework, as illustrated in Fig. 1. Based on this systematic optimization, the final recipe was established at an input power of 2000 W, a chamber pressure of 4 mTorr, and a gas flow rate of 30 sccm Ar and 70 sccm CH_4 [9]. To investigate the influence of the a-C:H layer thickness on bonding characteristics, the plasma processing time was systematically varied at 15, 30, 45, and 60 seconds, with 60 seconds being identified as the primary condition for the optimized recipe. The chemical states and elemental distribution of the plasma-treated Cu surface were characterized using X-ray photoelectron spectroscopy (XPS) and Time-of-flight secondary ion mass spectrometry (ToF-SIMS). Furthermore, the cross-sectional morphology and thickness of the deposited a-C:H layer were verified via Transmission electron microscopy (TEM).

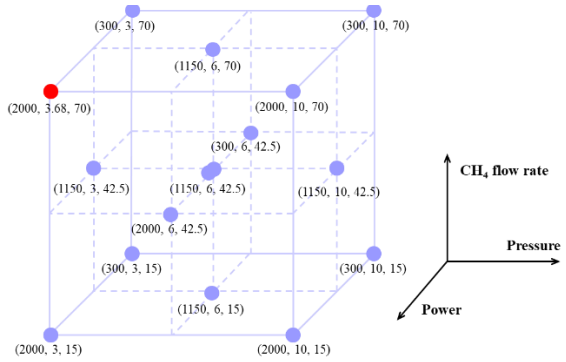


Fig. 1 Central composite design (CCD) of plasma treatment parameters for process optimization.

B. Wafer Bonding Procedure

To evaluate the interfacial integrity, Cu-to-Cu direct wafer bonding was performed in a high-vacuum environment. The plasma-treated wafers were aligned and bonded at 260 °C for 1 hour. Following the initial bonding process, a post-bonding annealing (PBA) step was conducted at 200 °C for an additional hour. After bonding, the quality and microstructure of the bonding interface were rigorously analyzed. Field-emission scanning electron microscopy (FE-SEM) was utilized to inspect the cross-sectional bonded interface for any micro-voids or delamination. To further investigate the atomic-scale metallic interconnectivity and Cu grain growth across the interface, cross sectional transmission electron microscopy (Cs-TEM) was employed. This allowed for an in-depth observation of the epitaxial-like grain growth and the behavior of the residual a-C:H layer at the bonding seam.

III. Result and Discussion

A. Surface Chemical Composition and Reduction Behavior

Fig. 2(a) presents the XPS wide scan spectra of the Cu surfaces before and after the Ar-CH_4 plasma treatment. Compared to the non-treated (pristine) Cu surface, all plasma-treated samples exhibited a significant reduction in the O1s peak intensity, accompanied by a sharp increase in the C1s peak. This observation clearly indicates that the Ar-CH_4 plasma effectively reduces the native Cu oxides through hydrogen radicals while simultaneously depositing a carbonaceous passivation layer (a-C:H) on the surface. Notably, the Cu2p peak intensity showed an initial increase at P15 but gradually decreased as the treatment time progressed to P60. This attenuation is attributed to the increasing thickness of the a-C:H layer, which limits the escape depth of the Cu photoelectrons from the underlying metal surface. The high-resolution spectra of the Cu2p region (Fig. 2(b)) further support these findings. The distinct Cu^{2+} satellite peaks observed in the non-treated sample completely disappeared after plasma treatment, indicating the successful removal of Cu^{2+}-related oxide species even at the initial stages of the process. This single-step reduction and passivation strategy effectively prevents re-oxidation, ensuring a stable Cu surface for subsequent bonding.

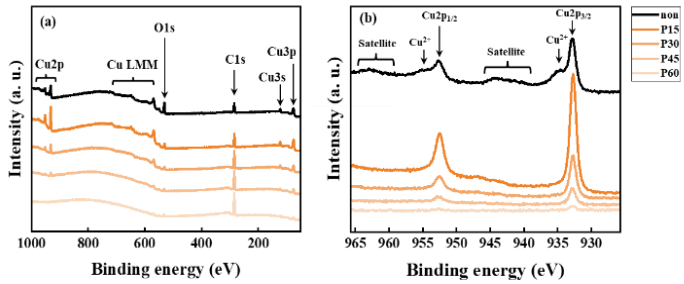


Fig. 2 XPS analysis of Cu surfaces from non-plasma through P60 treatment conditions: (a) wide scan spectra and (b) high-resolution Cu 2p spectra

The physical thickness and interfacial quality of the deposited a-C:H layers were further examined using cross-sectional TEM. To preserve the original state of the thin carbonaceous layer, a sacrificial metal capping layer was deposited onto the a-C:H surface. This layer serves to protect the surface from potential structural damage during the Focused Ion Beam (FIB) milling process required for TEM lamella preparation. As shown in the images, the left-hand side represents the plasma-treated Cu film, while the right-hand side is the sacrificial capping layer. Fig. 3(a)-(e) displays the Cs-TEM images of the Cu surface from the non-plasma condition through P60. While the non-treated surface exhibited an

irregular native oxide layer, the plasma-treated samples showed a distinct, uniform amorphous layer (a-C:H) on the Cu film. As summarized in Fig. 3(f), the thickness of the a-C:H layer increased linearly with the plasma treatment time, demonstrating high controllability within the nanometer scale.Interestingly, in the P30 and P60 samples, a significantly darker region was observed at the interface between the Cu film and the a-C:H layer. In TEM imaging, darker contrast typically corresponds to regions containing heavier atoms or higher density. This localized contrast enhancement indicates that the light oxygen atoms from the native oxide were effectively removed and replaced by a dense, metallic Cu-rich environment through the hydrogen-radical-driven reduction process. This microstructural evidence, consistent with the previous XPS results, confirms that the CH_4 plasma treatment achieves effective surface reduction prior to the full development of the carbonaceous passivation barrier.

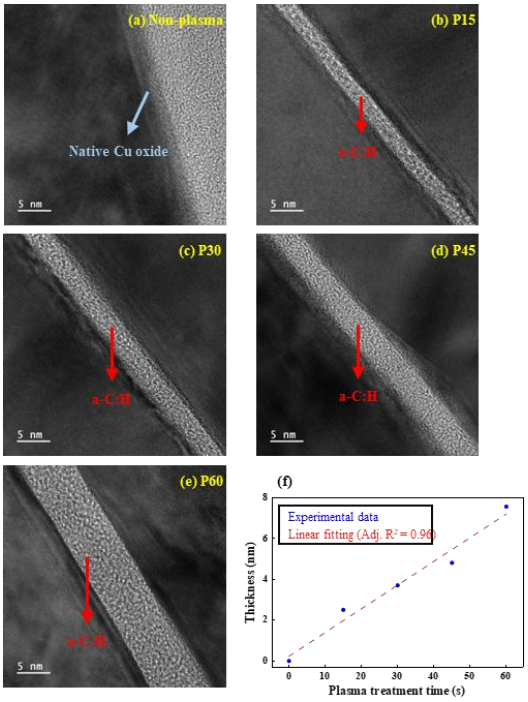


Fig. 3 Cross-sectional HR-TEM analysis of a-C:H layers on Cu films: (a)-(e) microstructural evolution from non-plasma through P60 conditions (left: Cu film, right: sacrificial capping layer), and (f) linear relationship between a-C:H thickness and plasma treatment time.

B. *Characterization of Bonded Interfaces and Diffusion Behavior*

The evolution of the bonding interface as a function of the initial a-C:H thickness is presented in Fig. 4. A comparative analysis of the cross-sectional TEM images (Fig. 4(a), (c), (e), (g)) clearly demonstrates that the interfacial morphology is highly dependent on the plasma-induced passivation layer. For the P15 condition, characterized by the thinnest a-C:H layer, the bonding seam is virtually eliminated, resulting in a monolithic Cu structure with continuous grain boundaries across the interface (Fig. 4(a)). This signifies that the carbonaceous layer at this thickness does not obstruct atomic movement, but rather facilitates extensive metallic interdiffusion. In contrast, as the treatment time increases toward P60, a persistent and distinct interlayer becomes visible, which physically segregates the two Cu surfaces and inhibits metallic contact.

The chemical transition at these interfaces was further quantified through EDS line profiles as shown in Fig. 4(b), (d), (f), (h). The data reveals a synchronized diffusion behavior between carbon and oxygen. In the P15 sample, the interfacial carbon concentration is significantly attenuated, indicating near-complete dissolution into the Cu matrix. This carbon diffusion appears to trigger a simultaneous inward migration of oxygen, preventing the localized accumulation of oxides at the bonding seam. However, a transition from diffusion-active to barrier-limited behavior occurs as the passivation layer thickens. In the P45 and P60 samples, the EDS profiles show sharp, localized peaks for both C and O at the interface (Fig. 4(f), (h)). At these regimes, the a-C:H layer remains thermally stable and maintains its amorphous integrity, thereby acting as a diffusion barrier that stifles the transport of both Cu and O atoms.

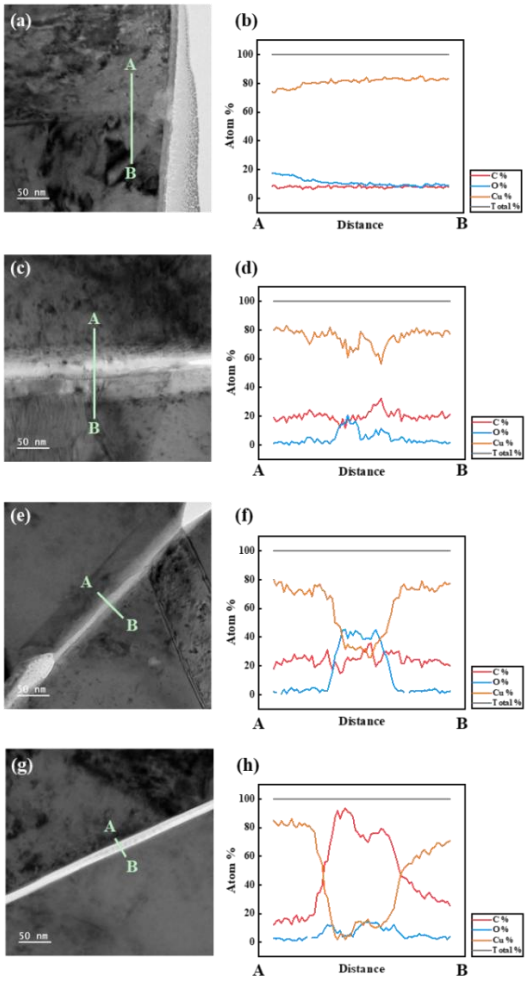


Fig. 4 Cross-sectional TEM images and corresponding EDS line profiles of the bonded Cu-Cu interfaces for different plasma treatment times: (a, b) P15, (c, d) P30, (e, f) P45, and (g, h) P60.

C. *Proposed Cu-C Interdiffusion Mechanism for Seamless Bonding*

Based on the cross-sectional TEM and EDS observations, the superior bonding quality of the P15 condition can be explained by a synergistic triple-diffusion mechanism that occurs during the thermal bonding process. As illustrated in Fig. 5, this mechanism allows the thin a-C:H passivation layer to transition from a surface protector to a diffusion-promoting interlayer. The process is governed by the following three pathways: (1) Cu diffusion through the a-C:H network: At the initial stage of bonding, Cu atoms migrate through intrinsic defects and voids within the carbonaceous clusters, initiating metallic contact at the nanometer scale. (2) Carbon migration along Cu grain boundaries: As the temperature increases, carbon atoms from the thin passivation layer preferentially migrate along the Cu grain boundaries. This redistribution facilitates interfacial rearrangement and enables the subsequent interdiffusion of Cu atoms across the original bonding seam. (3) Lattice-scale carbon diffusion: Simultaneously, a portion of the carbon atoms diffuses into the FCC copper lattice [10, 11]. This interstitial diffusion minimizes the presence of a discrete interfacial layer, resulting in the monolithic, seamless grain structure observed in the P15 TEM images.

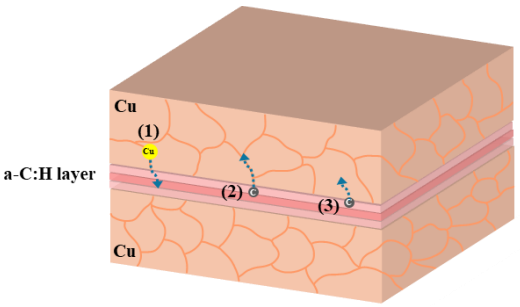


Fig. 5 Schematic illustration of the synergistic triple-diffusion mechanism in the P15 sample: (1) Cu diffusion through carbonaceous defects, (2) carbon migration along Cu grain boundaries, and (3) carbon diffusion into the Cu lattice.

IV. Conclusion

This study successfully demonstrated a highly efficient and scalable surface engineering strategy for next-generation Cu hybrid bonding through a single-step Ar/CH_4 plasma treatment. By utilizing an ICP-CVD system, we achieved the simultaneous reduction of native Cu oxides and the deposition of a nanometer-scale a-C:H passivation layer, effectively streamlining the pre-bonding process. The optimization of plasma parameters through a Central Composite Design (CCD) allowed for precise control over the carbonaceous layer's thickness, which was found to be the governing factor for the final bonding integrity.

Our findings reveal a critical transition in the role of the a-C:H layer depending on its thickness. While thicker layers act as a persistent diffusion barrier that obstructs metallic intermixing, the optimized P15 condition initiates a synergistic triple-diffusion mechanism during the thermal bonding process at 260°C. This mechanism involves the coordinated transport of Cu atoms through carbonaceous defects, the migration of carbon along Cu grain boundaries, and the eventual interstitial diffusion of carbon into the Cu lattice. The elemental redistribution observed via EDS line profiles further confirmed that this active carbon diffusion facilitates the inward migration of residual oxygen, preventing the accumulation of brittle oxides at the interface. As a result, the P15 sample achieved a seamless, monolithic Cu interface with continuous grain growth, effectively eliminating the bonding seam. This research provides a robust framework for balancing surface protection with metallurgical bonding requirements, offering a promising solution for fine-pitch interconnects in high-performance AI and HPC packaging architectures.

Acknowledgment

This work was supported by K-CHIPS (Korea Collaborative & High-tech Initiative for Prospective Semiconductor Research) (2410009462, RS-2023-00236091, 23035-45FC) funded by the Ministry of Trade. Industry & Energy (MOTIE, Korea).

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